SOC 690S: Machine Learning in Causal Inference

Week 10: Regression Discontinuity Design

Wenhao Jiang
Department of Sociology, Fall 2025

Week	Date	Topic	Problem sets	
			Assign	Due
1	Aug 26	Introduction: Motivation and Linear Regression		
2	Sep 2	Foundation: ML-based Regression Regularization		
3	Sep 9	Foundation: Non-Linear Machine Learning	1	
4	Sep 16	Foundation: Double Machine Learning		
5	Sep 23	Foundation: Directed Acyclic Graph (DAG)	2	1
6	Sep 29	Core: Matching, Propensity, Weighting, and Beyond		
7	Oct 7	Core: Instrumental Variable Estimation	3	2
8	Oct 14	*Fall break*		
9	Oct 21	**In-class midterm**		
10	Oct 28	Core: Regression Discontinuity Design		
11	Nov 4	Core: Panel Data and Difference-in-Difference		3
12	Nov 11	Advanced: Heterogeneous Treatment Effect I	4	
13	Nov 18†	Advanced: Heterogeneous Treatment Effect II		
14	Nov 25	Advanced: Unstructured Data Feature Engineering		4
	Dec 13	**final project due**		

† Justin Lucas Sola, Assistant Professor of Sociology and Data Science at UNC, will join us to give a guest talk on heterogeneous treatment effects and their applications.

Sharp Regression Discontinuity

Sharp Regression Discontinuity (RD): Setup

- In sharp RD, treatment status is a deterministic and discontinuous function a covariate x_i
- Suppose, for example, that

$$D_i = \begin{cases} 1 & \text{if } X_i \geq c \\ 0 & \text{if } X_i < c \end{cases}$$

  where c is a known threshold or cutoff
- The assignment mechanism is a deterministic function of X_i because once we know X_i we know D_i
- Treatment is a discontinuous function of X_i because no matter how close X_i gets to c, treatment is unchanged until $X_i = c$

Sharp Regression Discontinuity (RD): Assumptions

- There are two necessary assumptions for RD identification
- 1. Local regularity, or continuity of potential outcomes

$$\lim_{x \downarrow c} E[Y_i(0)|X_i = x] = \lim_{x \uparrow c} E[Y_i(0)|X_i = x]$$
$$\lim_{x \downarrow c} E[Y_i(1)|X_i = x] = \lim_{x \uparrow c} E[Y_i(1)|X_i = x]$$

- It implies that the potential outcomes evolve smoothly in X_i; there are no unobserved characteristics that discontinuously change at c other than X_i
- 2. No manipulation of the running variable: individuals cannot sort or manipulate X_i around c; the density $f(X_i)$ is continuous at the cutoff

$$\lim_{x \downarrow c} f(x_i) = \lim_{x \uparrow c} f(x_i)$$

- Both can be tested using McCracy (2008) (in `rdd:DCdensity()` in R)

Sharp Regression Discontinuity (RD): Setup

The parameter of interest in RDD is the local ATE at the cutoff value c

$$\tau = E[Y_i(1) - Y_i(0) | X_i = c]$$

Under local regularity

$$\tau = \lim_{x \downarrow c} E[Y_i | X_i = x] - \lim_{x \uparrow c} E[Y_i | X_i = x]$$
$$= g_+(c) - g_-(c)$$

when we express CEF by $g(\cdot)$

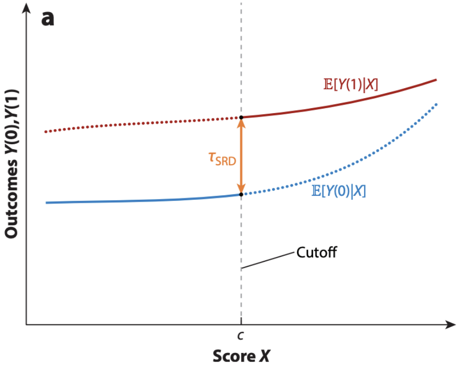


Figure: Sharp RDD setup

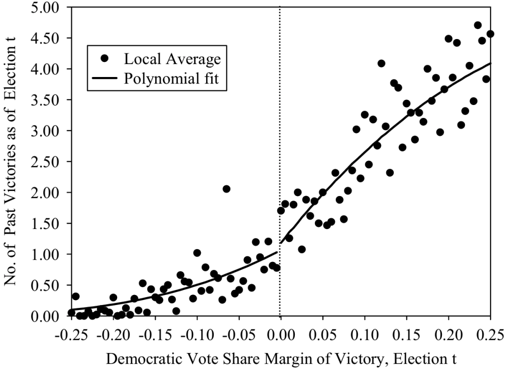
Sharp Regression Discontinuity (RD): Example

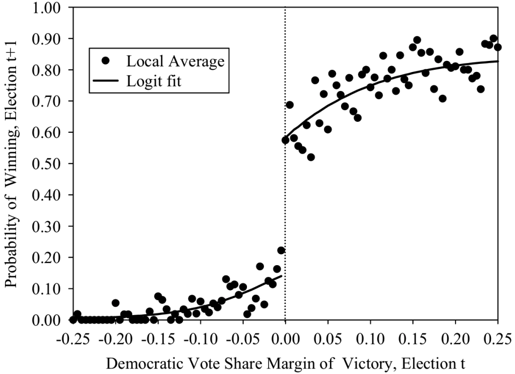
- One question in social science is whether receiving academic recognition in early career has long-term positive effects
- Does receiving finalist status or eligibility for a merit-based scholarship affects students' later educational outcomes—such as college attendance, college quality, or academic achievement
- PSAT score determines the eligibility for National Merit Scholarship
- Students just above and below the PSAT threshold c are nearly identical in ability; any discontinuity in $E[Y_i \mid X_i]$ at c is interpreted as the causal effect of eligibility (Campbell 1969)

Sharp Regression Discontinuity (RD): Lee (2008)

- Lee (2008) examined the causal effect of incumbency on electoral success in U.S. House races
- Focused on elections that are extremely close, where two candidates—typically a Democrat and a Republican—receive nearly the same share of the vote
- In these very tight contests, the outcome is determined by essentially random factors such as last-minute events, turnout fluctuations, or counting noise
- Candidates who barely win and those who barely lose are, on average, comparable in ability, popularity, and campaign resources

Regression Discontinuity (RD): Lee (2008)





Regression Discontinuity (RD): Lei and Zhou (2022)

- Lei and Zhou (2022) explored why Chinese local governments invest heavily in long-term infrastructure projects—especially in situations where a local leader's tenure is short and the project pay-offs extend far into the future
- Local officials can still gain private political returns by initiating high-visibility infrastructure projects that elevate the economic performance of their jurisdiction
- To be eligible for creating a subway system, a city must satisfy four central government requirements (1) the city's annual fiscal revenue exceeds 10 billion, (2) the city's GDP reaches at least 100 billion,(3) the city's population exceeds 3 million people, and (4) more than 30,000 people per hour are expected to use a subway line
- While cities satisfying these four requirements are not guaranteed to have subways, those not satisfying them are not eligible

Sharp Regression Discontinuity (RD): Local Fit

- An important feature of RD is that there is *no* value of X_i at which we get to observe both treatment and control observations
- Unlike full covariate matching strategies, which are based on treatment-control comparisons conditional on covariate values where there is some overlap, the validity of RD turns on our willingness to extrapolate across covariate values, at least in a neighborhood of the discontinuity
- The functional form of the regression becomes essential for estimation

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- The functional form of the regression becomes essential for estimation
- The most common functional form of Sharp RD is a (local) linear regression

$$\begin{aligned} Y_i &= a_+ + b_+(x_i - c) + e_i && \text{for } x_i \geq c \\ Y_i &= a_- + b_-(x_i - c) + e_i && \text{for } x_i < c \\ \hat{\tau} &= a_+ - a_- \end{aligned}$$

Bandwidth Selection Intuition

- Given the estimated ATE is local to the *cutoff*, ideally the bandwidth h used in local linear regression should be reasonably small
- To avoid introducing large *biases* from including observations further away from the cutoff that may be systematically different from observations at the cutoff in unobserved ways (therefore distorting the prediction of a_+ and a_-)
- But it cannot be too small, as otherwise very few observations will be used; estimation uncertainty (variance) would be large
- This is the classic bias–variance tradeoff

Local Kernel Weight

- Besides tuning the bandwidth, a second (and complementary) way to reduce bias from distant observations is to apply *local kernel weights*
- Replace raw weights by $K(u)$ so observations farther from the cutoff get less influence, where $u = (x - c)/h$
- Common kernels are

$$\begin{cases} K(u) & = (1 - |u|) \cdot \mathbb{1}(|u| \leq 1) \quad \text{triangular} \\ K(u) & = \frac{3}{4}(1 - u^2) \cdot \mathbb{1}(|u| \leq 1) \quad \text{Epanechnikov} \\ K(u) & = \frac{1}{\sqrt{2\pi}} \exp\Big(- \frac{u^2}{2} \Big) \quad \text{Gaussian} \end{cases}$$

- *Triangular* kernel is often recommended in RDD practice; it places greater weight on observations closest to the cutoff and reduces boundary bias (the default in many RDD packages)
- Choice of kernel is much less important than choice of bandwidth

Local Regression Form

- For the same reasons, high-order polynomials are not recommended as they may introduce significant biases in extrapolation and prediction at cutoff c if the true DGP does *not* have high-order terms

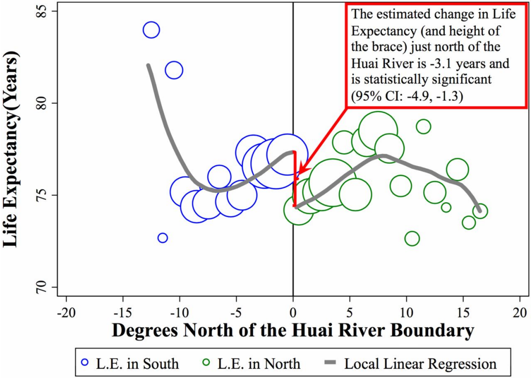

Figure: Ebenstein et al. (2017)

Local Regression Form

- If there are sufficient number of observations around the cutoff c, a more natural estimand is a local nonparametric mean difference

$$
\begin{aligned}
E[Y_i(0) \mid X_i = c] &\simeq E[Y_i \mid c - \Delta < X_i < c] \\
E[Y_i(1) \mid X_i = c] &\simeq E[Y_i \mid c \leq X_i < c + \Delta]
\end{aligned}
$$

$$
\lim_{\Delta \to 0} E[Y_i \mid c \leq X_i < c + \Delta] - E[Y_i \mid c - \Delta < X_i < c] = E[Y_i(1) - Y_i(0) | X_i = c]
$$

- The bandwidth h is chosen as the largest local window where local randomization assumption (covariate balance) is met (Cattaneo, Frandsen, and Titiunik 2015)

Bias and Variance Tradeoff in Bandwidth Selection

- Recall that we define right and left limits

$$g_+(c) = \lim_{x \downarrow c} g(x), \quad g_-(c) = \lim_{x \uparrow c} g(x)$$

- And we estimate the treatment effect at the cutoff

$$\tau = g_+(c) - g_-(c)$$

by separate *one-sided* local linear regressions using bandwidth h

- Focusing on the right hand side of the cutoff with a kernel function $K(\cdot)$

$$(\hat{a}_+, \hat{b}_+) = \arg\min_{a,b} \sum_{i;X_i>c} K\left(\frac{X_i - c}{h}\right)[Y_i - a - b(x_i - c)]^2$$

Bias and Variance Tradeoff in Bandwidth Selection

- The mean squared error (MSE) of $\hat{a}_+$ can be decomposed as
$$E\left[\left(\hat{a}_+ - g_+(c)\right)^2\right] = \left(E[\hat{a}_+] - g_+(c)\right)^2 + V(\hat{a}_+)$$
- The optimal bandwidth h minimizes the MSE

Bias and Variance Tradeoff in Bandwidth Selection

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$$E\left[(\hat{a}_+ - g_+(c))^2\right] = \left(E[\hat{a}_+] - g_+(c)\right)^2 + V(\hat{a}_+)$$
- The optimal bandwidth h minimizes the MSE
- To make notations a bit simpler, define
$$u_i = \frac{X_i - c}{h} \quad \text{such that } u_i \in [0, 1]$$
- Expand Y_i at cutoff c using Taylor expansion
$$\begin{aligned} Y_i &= g_+(X_i) + e_i \\ &= g_+(c) + g'_+(c)(X_i - c) + \frac{1}{2} g''_+(c)(X_i - c)^2 + \mathcal{O}_3 + e_i \\ &= g_+(c) + g'_+(c) h u_i + \frac{1}{2} g''_+(c) h^2 u_i^2 + e_i \end{aligned}$$

Bias and Variance Tradeoff in Bandwidth Selection

- Solving the *population* normal equation for the intercept a_+

$$
\begin{aligned}
0 &= E\left[K(u_i)(Y_i - a_+ - b_+ h u_i)\right] \\
&= E\left[K(u_i)\left(g_+(c) + g'_+(c) h u_i + \frac{1}{2} g''_+(c) h^2 u_i^2 - a_+ - b_+ h u_i\right)\right] \\
&= \int_0^1 K(u)\left(g_+(c) + g'_+(c) h u + \frac{1}{2} g''_+(c) h^2 u^2 - a_+ - b_+ h u\right) du \\
&= (g_+(c) - a_+)\int_0^1 K(u) d_u + (g'_+(c) - b_+) h \int_0^1 u K(u) du + \\
&\qquad \frac{1}{2} g''_+(c) h^2 \int_0^1 u^2 K(u) du
\end{aligned}
$$

Bias and Variance Tradeoff in Bandwidth Selection

- Define the kernel moments

$$
\begin{aligned}
m_j &= \int_0^1 u^j K(u)du, \quad j = 0, 1, 2 \\
0 &= (g_+(c) - a_+)m_0 + (g'_+(c) - b_+)hm_1 + \frac{1}{2}g''_+(c)h^2 m_2 \\
&= (g_+(c) - a_+) + (g'_+(c) - b_+)h\frac{m_1}{m_0} + \frac{1}{2}g''_+(c)h^2\frac{m_2}{m_0} \\
&= (g_+(c) - a_+) + (g'_+(c) - b_+)h\mu_1 + \frac{1}{2}g''_+(c)h^2 h\mu_2
\end{aligned}
$$

where

$$
\mu_j = \frac{m_j}{m_0}
$$

Bias and Variance Tradeoff in Bandwidth Selection

- Define the *bias* of the estimated a_+

$$\Delta(\hat{a}_+) = \hat{a}_+ - g_+(c)$$

- We can similarly solve the *population* normal equation for the slope b_+, with

$$\Delta(\hat{b}_+) = h\left(\hat{b}_+ - g_+(c)\right)$$

- Solving the two normal equations

$$\Delta(\hat{a}_+) + \mu_1 \Delta(\hat{b}_+) = -\frac{1}{2}h^2 g''_+(c)\mu_2$$
$$\mu_1 \Delta(\hat{a}_+) + \mu_2 \Delta(\hat{b}_+) = -\frac{1}{2}h^2 g''_+(c)\mu_3$$

Bias and Variance Tradeoff in Bandwidth Selection

- Simplifying the two normal equations, we will derive

$$\begin{aligned}\Delta(\hat{a}_+) &= \frac{h^2}{2} g''_+(c) \frac{\mu_1 \mu_3 - \mu_2^2}{\mu_2 - \mu_1^2} \\ &= \frac{h^2}{2} g''_+(c) A_k\end{aligned}$$

Bias and Variance Tradeoff in Bandwidth Selection

- In more compact matrix notation

$$\Delta(\hat{a}_+) = v_1' \left(E[K(u_i)X_iX_i']\right)^{-1} E[K(u_i)X_ie_i]$$
$$V(\hat{a}_+) = v_1' \left(E[K(u_i)X_iX_i']\right)^{-1} V\left(E[K(u_i)X_ie_i]\right) \left(E[K(u_i)X_iX_i']\right)^{-1} v_1$$

  where $v_1 = (1, 0)$

- The intuition is from the standard OLS, where

$$\hat{\beta} - \beta = (X'X)^{-1}X'e$$

Bias and Variance Tradeoff in Bandwidth Selection

- Suppose u_i has density approximately $f(c)$ near c

$$E[K(u_i)X_iX_i'] = f(c)\int_0^1 K(u)\begin{pmatrix}1\\u\end{pmatrix}(1,u)du = f(c)\begin{pmatrix}m_0 & m_1\\ m_1 & m_2\end{pmatrix}$$

- And the "meat"

$$\begin{aligned}V(E[K(u_i)X_ie_i]) &= E[K(u_i)^2X_iX_i'e_i^2]\\ &= \sigma_+^2 f(c)\int_0^1 K(u)^2\begin{pmatrix}1\\u\end{pmatrix}(1,u)du\\ &= \sigma_+^2 f(c)\begin{pmatrix}r_0 & r_1\\ r_1 & r_2\end{pmatrix}\end{aligned}$$

where

$$\sigma_+^2 = V(e_i \mid X_i) \text{ and } r_j = \int_0^1 u^j K(u)^2 du$$

Bias and Variance Tradeoff in Bandwidth Selection

- Simplifying terms, we will eventually derive defined as

$$
\begin{aligned}
V(\hat{a}_+) &= \frac{\sigma_+^2}{nhf(c)} \frac{m_2^2 r_0 - 2m_1 m_2 r_1 + m_1^2 r^2}{(m_0 m_2 - m_1^2)^2} \\
&= \frac{\sigma_+^2}{nhf(c)} B_K
\end{aligned}
$$

where $nhf(c)$ is the approximate effective size in window $[c, c+h]$

The Imbens–Kalyanaraman Bandwidth

- For the estimated treatment effect $\hat{\tau}$, the sum of mean squared error can be decomposed as the sum of the square of bias and variance

$$\Delta(\hat{\tau})^2 + V(\hat{\tau}) = \frac{A_K^2}{4}\left(g''_+(c) - g''_-(c)\right)^2 h^4 + \frac{B_K}{nhf(c)}\left(\sigma_+^2 - \sigma_-^2\right)$$

- Solving FOC *w.r.t.* h to minimize the sum and to strike a balance between variance and bias, we have

$$h_{\text{IK}} = \left[\frac{B_K}{A_K^2}\frac{\sigma_+^2 + \sigma_-^2}{nf(c)\,\left(g''_+(c) - g''_-(c)\right)^2}\right]^{1/5}$$

Covariates Controls in Regression Discontinuity

Motivations of Including Covariates

- The most important benefits of including covariates is to improve estimation efficiency
- As in RCTs, using covariates can increase efficiency and improve power
- The intuition is that the variance of the estimand is a function of σ^2, the variance of the error term unexplained by X_i (see a formal derivation of the quantity on page 475 in Chapter 17 of CML)
- Importantly, the estimation bias is *not* a function of covariates

Including Covariates in Sharp RD

- In the regular RDD without covariates, the compact approach to write the estimand of ATE is

$$\hat{\tau}(h) = v_2' \arg\min_{\theta \in \mathbb{R}^4} \sum_{i=1}^{n} K_h(X_i - c)\left(Y_i - \theta' V_i\right)^2$$

- $V_i = (1, D_i, (X_i - c), D_i(X_i - c))'$
- $K_h(x)$ is kernel weights around c
- h = bandwidth
- $v_2 = (0, 1, 0, 0)'$ selects coefficient on D_i

Including Covariates in Sharp RD

- With covariates, the compact approach to write the estimand of ATE is

$$\hat{\tau}(h) = v_2' \arg \min_{\theta,\gamma \in \mathbb{R}^{(4+p)}} \sum_{i=1}^{n} K_h(X_i - c) \left(Y_i - \theta' V_i - \gamma' Z_i \right)^2$$

- This will be equivalent to the residualized version

$$\hat{\tau}(h) = v_2' \arg \min_{\theta \in \mathbb{R}^4} \sum_{i=1}^{n} K_h(X_i - c) \left(\tilde{Y}_i - \theta' V_i \right)^2$$

- where

$$\tilde{Y}_i = Y_i - Z_i' \hat{\gamma}$$

$$\hat{\gamma} = \arg \min_{\gamma} \sum_{i=1}^{n} K_h(X_i - c) \left(Y_i - \gamma' Z_i \right)^2$$

Including Covariates in Sharp RD

- As before, the estimation of γ does not have to be linear or in low dimension
- One may flexibly estimate the nuisance function $\eta_0(Z_i) = E[Y_i \mid Z_i]$ using ML models and define the covariate-adjusted ATE as

$$\tau = \lim_{x \downarrow c} E[Y_i - \eta_0(Z_i)|X_i = x] - \lim_{x \uparrow c} E[Y_i - \eta_0(Z_i)|X_i = x]$$

Fuzzy Regression Discontinuity

Fuzzy Regression Discontinuity (RD): Basic Setup

- Fuzzy RD exploits discontinuities in the *probability* or *expected value* of treatment conditional on a covariate
- The discontinuity becomes an instrumental variable for treatment status instead of deterministically switching treatment on or off

$$P(D_i = 1 \mid x_i) = \begin{cases} g_1(x_i) & \text{if } x_i \geq c \\ g_0(x_i) & \text{if } x_i < c \end{cases}$$

  where $g_0(c) \neq g_1(c)$

- The simple difference $\tau = \lim_{x \downarrow c} E[Y_i \mid X_i = x] - \lim_{x \uparrow c} E[Y_i \mid X_i = x]$ as in sharp RD does not capture local ATE because there is a mixture of treated and control units on both sides

Fuzzy Regression Discontinuity (RD): Basic Setup

- However, the local ATE in the fuzzy RD does not systematically differ
- It is intuitive to show that

$$\tau = \frac{\lim_{x \downarrow c} E[Y_i \mid X_i = x] - \lim_{x \uparrow c} E[Y_i \mid X_i = x]}{\lim_{x \downarrow c} E[D_i \mid X_i = x] - \lim_{x \uparrow c} E[D_i \mid X_i = x]}$$

- Under local regularity (continuity of potential outcome) and no manipulation of the running variable

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- Under local regularity (continuity of potential outcome) and no manipulation of the running variable
- At the cutoff, this is essentially the Wald estimator, where the side of X_i relative to c is the instrument
- The interpretation is the same as Wald estimator: the estimated ATE from fuzzy RD is the *local* ATE for *compliers* (Hahn, Todd, and van der Klaauw 2001)

Fuzzy Regression Discontinuity (RD): Estimation

- We apply the standard sharp RD specification on both denominator and numerator; we define $Z_i = \mathbb{1}(X_i \geq c)$
- The numerator can be estimated by a local linear regression

$$Y_i = \alpha_Y + \beta_Y Z_i + \gamma_Y (X_i - c) + \eta_Y Z_i (X_i - c) + \epsilon_i$$

  where the estimated discontinuity of the numerator is $\hat{\beta}_Y$
- The denominator symmetrically can be estimated by

$$D_i = \alpha_D + \beta_D Z_i + \gamma_D (X_i - c) + \eta_D Z_i (X_i - c) + \mu_i$$

  where the estimated discontinuity of the denominator is $\hat{\beta}_D$
- The ATE is defined as $\hat{\tau} = \hat{\beta}_Y / \hat{\beta}_D$
- Local kernel weight $K(\cdot)$ can be applied

Fuzzy Regression Discontinuity (RD): 2SLS Estimation

- Not surprisingly, this *reduced-form* of estimation is numerically identical as 2SLS
- Remember we are concerned about endogeneity of treatment D_i on Y_i
- But around the cutoff, the (probability of) assignment of D_i can be viewed as random
- Therefore, $Z_i = \mathbb{1}(X_i \geq c)$ may be used as an instrument for D_i around c within a reasonable bandwidth h (Cattaneo et al. 2020)
  - Z_i significantly predicts D_i
  - Z_i is independent of the final outcome except through D_i

Fuzzy Regression Discontinuity (RD): 2SLS Estimation

- The local structural (second-stage) equation of interest is

$$Y_i = \gamma_0 + \gamma_1 D_i + \gamma_2(X_i - c) + \gamma_3 D_i(X_i - c) + \epsilon_i$$

where D_i and $D_i(X_i - c)$ are *endogenous*

Fuzzy Regression Discontinuity (RD): 2SLS Estimation

- The local structural (second-stage) equation of interest is

$$Y_i = \gamma_0 + \gamma_1 D_i + \gamma_2 (X_i - c) + \gamma_3 D_i (X_i - c) + \epsilon_i$$

  where D_i and $D_i(X_i - c)$ are *endogenous*
- The cutoff indicator and its interaction serve as instruments

$$Z_i = \mathbb{1}(X_i \geq c) \text{ for } D_i \quad Z_i(X_i - c) \text{ for } D_i(X_i - c)$$

- Obtain fitted values from the first stage and plug them into the structural equation above for the second stage
- The estimated coefficient $\widehat{\gamma}_1$ is the local ATE at the cutoff
- Higher-order local polynomials follow the same one-to-one correspondence between regressors and instruments

Fuzzy RD: Dell and Querubin (2018)

- In this paper "*Nation Building Through Foreign Intervention: Evidence from Discontinuities in Military Strategies*" (in supplementary reading list)
- Dell and Querubín examine the effects of U.S. military intervention strategies during the Vietnam War by exploiting two types of discontinuities in how strategies were applied
- First, they exploit a rounding threshold in a Bayesian hamlet-security algorithm to identify variation in bombing intensity (top-down fire-power)
- Second, they exploit a geographical discontinuity between two adjacent military regions with differing counterinsurgency philosophies (one emphasizing "overwhelming firepower" and the other a "hearts and minds" bottom-up approach)

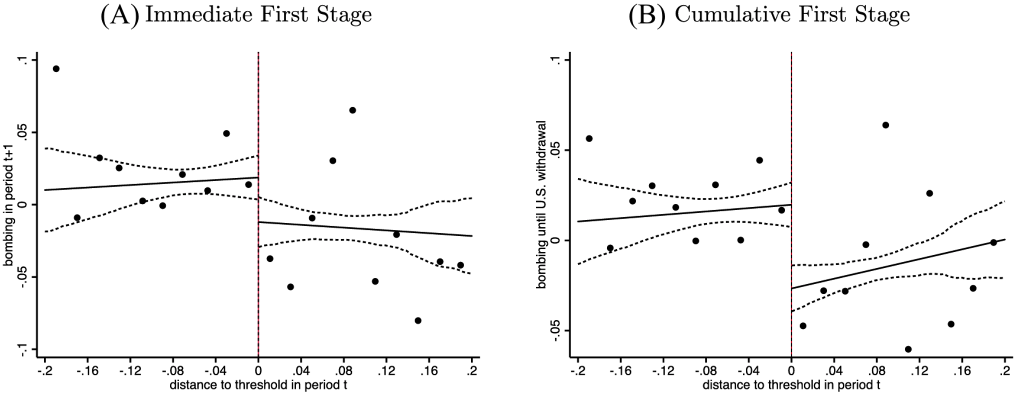
Fuzzy RD: Dell and Querubin (2018)

- The first-stage regression takes the following form (pp. 724)

$$y_{h,t+n} = \gamma_1 below_{ht} + \sum_{d=1}^{4} \delta_d D_{htd} + \sum_{d=1}^{4} \nu_d D_{htd} f_d(dist_{ht}) + \sum_{d=1}^{4} \psi_d D_{htd} f_d(dist_{ht}) below_{ht} + \alpha_t + \beta X_{ht} + \epsilon_{ht}$$

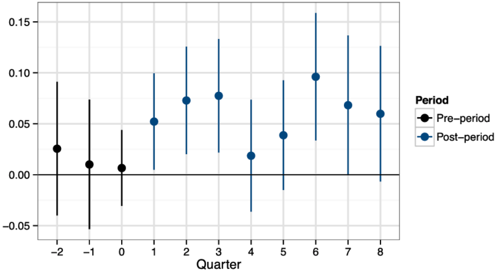
  - $y_{h,t+n}$ is bombing in hamlet h, in quarter(s) $t+n$
  - $below_{ht}$ is an indicator equal to 1 if the hamlet is below the threshold
  - $f_d(dist_{ht})$ is an RD polynomial in distance to the nearest threshold, estimated separately on either side of each threshold ($A\breve{}B$, $B\breve{}C$, $C\breve{}D$, $D\breve{}E$)
  - D_{htd} is a set of dummies equal to 1 if threshold d is the nearest threshold
  - X_{ht} are controls from question responses
  - α_t is a quarter-year fixed effect

Fuzzy RD First Stage: Dell and Querubin (2018)

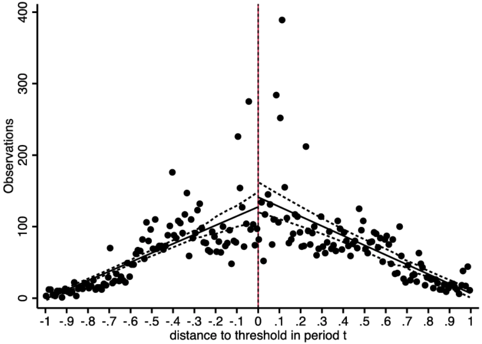


Fuzzy RD First Stage: Dell and Querubin (2018)

(C) Impacts by Quarter

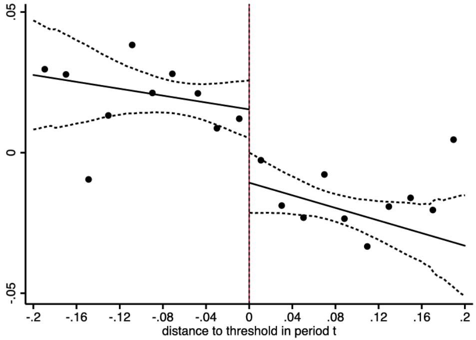


(D) McCrary Test

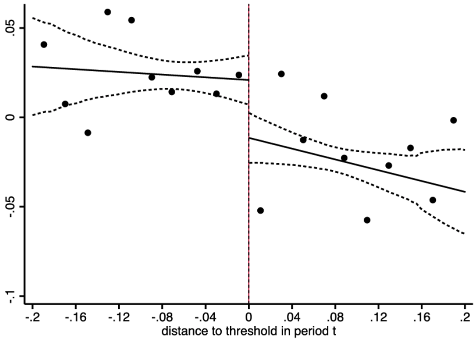


Fuzzy RD Reduced Form: Dell and Querubin (2018)

(A) *VC Presence (Cumulative)*



(B) *Active VC Infrastructure (Cumulative)*



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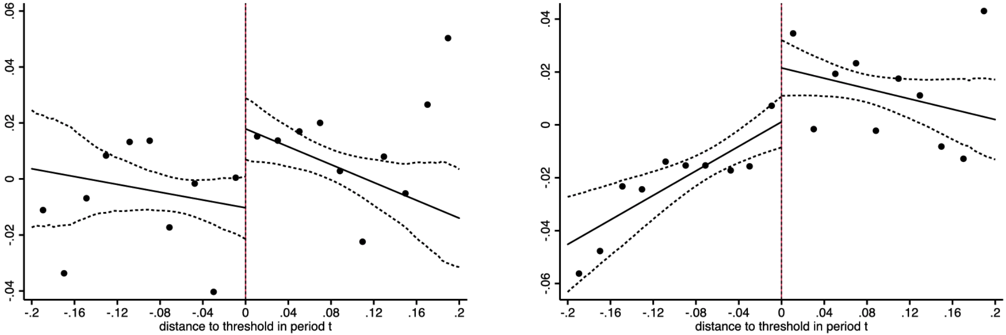
(E) *Access to a Primary School (Cumulative)*

(F) *Participation in Civic Orgs (Cumulative)*



Regression Discontinuity in Time

Regression Discontinuity in Time (RDiT): Setup

- RDiT is a special case of the regression discontinuity (RD) design where *time* is the running (forcing) variable
- At a known cutoff time T^*, a policy or intervention begins; observations just before and just after T^* are compared
- Key identifying assumptions are
  - Local regularity or continuity of potential outcome: units just before and just after T^* are comparable (unobserved time-varying confounders evolve smoothly)
  - No manipulation: there is no anticipation of the event for some groups
  - No other confounding event occurs exactly at T^*

RDiT: Estimation and Model Specification

- The local polynomial regression around the cutoff can be written as

$$Y_t = \alpha_0 + \alpha_1(T_t - T^*) + \tau D_t + \alpha_2(T_t - T^*)\, D_t + \epsilon_t, \quad \text{for } |T_t - T^*| < h$$

- Y_t is outcome at time t
- T_t is time index (running variable)
- $D_t = \mathbb{1}(T_t \geq T^*)$
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- h is bandwidth
- Control variables can be included in the same way as in standard RD

RDiT: Strengths and Limitations

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- The effect is estimated using observations close to the cutoff date T^* and the effect is essentially local ATE
- It captures only a *short-term, local change* in the outcome
- However, RDiT improves upon simple pre/post comparisons; unlike naive before-and-after analyses, it allows for flexible controls for smooth time trends and other continuous covariates
- This flexibility helps isolate the discrete policy effect at the exact intervention date
- All units receive treatment at the same time, leaving no untreated control group for comparison as in Difference-in-Difference